

Answers

1. **C.** With progressive pain and swelling, no history of trauma, low grade fever, and a warm, swollen, erythematous joint, arthrocentesis is indicated to rule out a septic joint.
2. **A.** Calcium is the most difficult vitamin to consume in sufficient quantities without eating meat or dairy.
3. **C.** Having the facility fax rather than mail results changes the method of delivery, but ultimately not the system in any meaningful way. The patient should follow up, but lab results indicating a serious illness should not wait to be tracked down at a patient follow up visit, which could be weeks from the initial visit, referring all patients with GI problems to specialists simply because the office cannot coordinate receiving lab results in a timely fashion is absurd, retraining the office staff implies a training or procedural error, when in fact this was likely a larger systemic error. Creating a log of tests needing to be followed up on creates a system where lab results are less likely to be missed.
4. **C.** Red flags for a compression fracture due to osteoporosis include prolonged corticosteroid use, never having received hormone replacement therapy following menopause, nonradiating point tenderness over a specific vertebral body, and an injury during a light task ("changing bedsheets") where an injury would not be expected.
5. **A.** The question is not asking which therapy you should initiate. It is asking which will be most beneficial. In reality you would advocate for smoking cessation for his overall health, you would add lisinopril for his hypertension, a high intensity statin is indicated because he is a diabetic with an ASCVD 10 year risk of 15.8% (i.e. $\geq 7.5\%$), you could increase his dose of HCTZ depending on how high the current dose is, and likewise you could increase his dose of metformin if he is not currently on a maximal dose. However, what will be most beneficial for the patient of the options listed is smoking cessation.
6. **A.** This is Tinea capitis. It commonly occurs in children. Black dot tinea capitis is common in African Americans. Diagnosis is confirmed with potassium hydroxide. Treat with oral griseofulvin for the patient only (not close contacts).
7. **A.** Randomized controlled trials have found that carotid endarterectomy (CEA) is beneficial for selected patients with asymptomatic internal carotid artery stenosis of 60 to 99 percent. However, the degree of benefit is not as great as it is for symptomatic carotid stenosis. This patient is asymptomatic and has only 30% stenosis in the worst carotid artery.
8. **A.** This patient has COPD, as indicated by the purulent sputum, exertional dyspnea, wheezing, history of cigarette smoking, diminished breath sounds, scattered crackles, and flattened diaphragms on CXR (they are laying it on thick in terms of classic COPD symptoms!). Furthermore the question essentially rules out serious CHF by describing no pedal edema, no S3 heart sound, no JVD. No therapy is described for this patient, so the very first drug to use would be an inhaled anticholinergic (ipratropium or tiotropium) $\pm$ albuterol inhaler, neither of which are options. Inhaled fluticasone is a step-up treatment if initial inhaled anticholinergic $\pm$ albuterol does not work, cromolyn is not indicated for COPD, oral amoxicillin is indicated for previously healthy, appropriately immunized infants and preschool children with mild to moderate community acquired pneumonia suspected to be of bacterial origin. For older patients, empiric treatment for community acquired pneumonia is typically azithromycin because it covers *Mycoplasma pneumoniae*. In any case, this patient likely does not have pneumonia. In studies considering smoking cessation in COPD patients, lung function decreased at twice the rate in patients who continued smoking versus those who quit, quitting smoking provided benefit whenever the person quit, and continuing smoking or relapsing worsened lung function.

9. **D.** This patient who has a positive PPD and a negative CXR has latent tuberculosis and should be treated with Isoniazid supplemented with vitamin B₆. While a repeat PPD 1-3 weeks after initial negative test is indicated for those not routinely tested for tuberculosis, a positive test does not require a repeat test for confirmation.

10. **B.** Carpal tunnel syndrome (CTS) refers to paresthesias, hypesthesia, pain, or numbness of the thumb, index, and middle fingers, as a result of compression of the median nerve in the carpal tunnel. Affected patients often awake with burning, numbness, and tingling in the median nerve distribution, which is bilateral in 75 percent of cases. Patients commonly report shaking the hand to relieve the discomfort. The diagnosis is based upon presence of characteristic symptoms and objective findings and is similar to that in nonpregnant individuals.

CTS is relatively common during pregnancy, with an incidence of 2 to 35 percent. The increased prevalence in pregnant women is thought to be caused by pregnancy-related fluid retention leading to compression of the nerve in the carpal tunnel; hormonal changes affecting the musculoskeletal system may also play a role. Symptoms tend to occur during the last trimester, but can occur at any time. In most cases, they gradually resolve over a period of weeks to months after delivery; however, symptoms can be prolonged for several months in women who are breast feeding. Symptoms may recur in subsequent pregnancies.

Patients may receive benefit from splinting the wrist at night in a neutral position or slight extension. Wrist splints may need to be worn throughout the day in severe cases. Corticosteroid injection or surgery to release the flexor retinaculum is rarely indicated during pregnancy since the disease has a better prognosis than idiopathic CTS and often resolves postpartum (UTD).

11. **G.** This patient has a classic presentation of a varicocele, as indicated by the “bag of worms” feel to the veins, the positional increase in symptoms when standing (due to the effect of gravity), and the painless nature of the mass. Varicoceles nearly always occur on the left due to the testicular vein on that side draining into the left renal vein rather than directly into the IVC as on the right side.

12. **C.** For diabetic neuropathy, the first line treatment is either a TCA or Gabapentin, then if those don't work, you can use opioids, topical capsaicin, or lidocaine.

13. **F.** This woman likely has Graves’ disease as indicated by the increased uptake on RAIU scan, but certainly with elevated T₄, T₃, and a low TSH, she has hyperthyroidism.

14. **C.** The most important thing in this case is to foster the physician-patient relationship, which will be undermined by either scheduling a family conference or ordering a urine toxicology screen. Women younger than 30 are never screened for HPV because studies have shown that they routinely contract then clear the virus. Having used marijuana four times in one’s lifetime does not constitute substance abuse, thus referral to a psychologist for this purpose is not reasonable.

15. **E.** This person has shingles along the V₁ distribution and should be treated with oral valacyclovir.

16. **A.** This patient has CHF as indicated by her exertional dyspnea, bibasilar lung crackles, S3 heart sound, and pitting edema of the lower extremities. In CHF, reduced ejection fraction leads to decreased stimulation of baroreceptors, artificially indicating to the body a low intravascular body. The body responds by secreting ADH, which causes fluid retention without sodium retention, producing volume overload with hyponatremia. Sadly, the resultant volume overload exacerbates the CHF by increasing preload, thus setting up a vicious cycle.

17. **B.**

Monosymptomatic (isolated) enuresis	
Definition	Urinary incontinence in children age ≥ 5
Workup	• Urinalysis • Urologic imaging for children with significant daytime symptoms & history of recurrent UTI
Management	• Behavior modifications ◦ Avoid sugary/caffeinated beverages ◦ Void regularly during the day & immediately before bedtime ◦ Drink ample fluids in the morning & early afternoon; minimize fluid intake before bedtime ◦ Reward system (eg, "gold star" chart) • Enuresis alarm: 1st-line intervention when behavior modifications fail; best long-term outcome • Pharmacotherapy: Best for short-term improvement; high risk of relapse ◦ 1st-line: Desmopressin ◦ 2nd-line: Tricyclic antidepressants

UTI = urinary tract infection.

18. **C.** HTN resistant to treatment is most likely caused by renal artery stenosis in older men, by fibromuscular dysplasia in young females. He also has hypokalemia due to increased aldosterone secretion as a result of increased renin production.

19. **A.** This patient has no evidence of strep pharyngitis (no tonsillar exudates, coughing, no cervical lymphadenopathy, no fever), no evidence of acute sinusitis, no evidence of epiglottitis, no evidence of pneumonia, and thus he likely has a viral URI requiring supportive treatment only.

20. **C.** This patient has classic symptoms of otitis externa, or "swimmer's ear": history of swimming, erythema and edema of the auditory canal with discharge, production of pain via manipulation of pinna, and normal looking eardrums. A number of preventive measures have been recommended for prevention of otitis externa, including use of earplugs while swimming, use of hair dryers on the lowest settings and head tilting to remove water from the ear canal, and avoidance of self-cleaning or scratching the ear canal. Acetic acid 2% (Vosol) otic solutions are also used, either two drops twice daily or two to five drops after water exposure. However, no randomized trials have examined the effectiveness of any of these measures.

21. **A.** This patient has CHF, as indicated by his dyspnea, orthopnea, cephalization on CXR, and Kerley B lines. Although diuretics will lead to some symptomatic relief, the question asks which drug will improve the patient's survival. ACE inhibitors are the only drug that will reduce mortality and prolong survival in moderate to severe CHF.

22. **B.** SCC in the vaginal and perineal region is most commonly caused by HPV infection.

23. **G.** Acute rheumatic fever can cause mitral valve regurgitation early and mitral stenosis later in life. The patient has an S4 due to left ventricular hypertrophy best heard at apex with patient in left lateral decubitus position and high atrial pressure.

24. **A.** This patient has exercise-induced and allergen-induced asthma. His asthma is not well-controlled on an albuterol inhaler alone, thus a daily inhaled corticosteroid should be added to his regimen.

25. **F.** This is tinea versicolor, which is best treated with topical selenium sulfide.

26. **E.** Nicotine patches are not contraindicated in patients with angina pectoris, nor should the dosage be lowered below recommended levels. Nicotine gum has not been shown to be superior to nicotine patches, and in fact patients are more likely to become addicted to the gum than the patch. Physicians counseling patients to cut back on cigarette use is associated with greater cessation failure rates compared to urging them to quit altogether. Transdermal nicotine in combination with a behavior modification program is a very effective method for achieving tobacco cessation.

27. **A.** Cluster headache is characterized by attacks of severe orbital, supraorbital, or temporal pain, accompanied by autonomic phenomena and/or restless or agitation. The unilateral autonomic symptoms associated with cluster headache, such as ptosis, miosis, lacrimation, conjunctival injection, rhinorrhea, and nasal congestion, occur only during the pain attack and are ipsilateral to the pain. The stereotypical attacks may strike up to eight times a day and are relatively short-lived. Another clinical landmark of the cluster headache syndrome is the circadian rhythmicity of the attacks. Cluster headache is strictly unilateral, and the symptoms remain on the same side of the head during a single cluster attack. However, the symptoms can switch to the other side during a different cluster attack (so-called side shift) in approximately 15 percent of cases. In contrast to migraines, patients with cluster headaches are restless and prefer to pace about or sit and rock back and forth. The attacks of cluster headache can be so vicious that patients may commit suicide if the disease is not diagnosed or treated.

28. **C.** This is not a good question. This patient is having an acute gout attack. Corticosteroids (e.g. prednisone, prednisolone) and NSAIDs (e.g. indomethacin or naproxen) are both reasonable first line agents for treatment of acute gout. Colchicine is a reasonable second line agent in patients with relative or absolute contraindications to both NSAIDs and corticosteroids, and in those in whom colchicine has effectively treated a gout flare previously. Historically, urate-lowering medications were thought to worsen acute gout flares, but recent evidence suggests that allopurinol (Zyloprim) can be started during an acute flare if it is used in conjunction with an NSAID and colchicine. Patients receiving a urate-lowering medication should be treated concurrently with an NSAID, colchicine, or low-dose corticosteroid to prevent a flare.

29. **A.** The question asks which one will have prevented mildly elevated AST/ALT. Of the answer choices only alcohol causes mildly elevated AST more than ALT. There is no carrier or chronic state of hepatitis A, so this asymptomatic patient likely does not have hepatitis. There is no reason to suspect that this patient has hemochromatosis, diphtheria, or tetanus. Smoking does not directly affect liver function or produce a rise in liver enzymes, and ALT is typically elevated more than AST in nonalcoholic fatty liver disease, which must be what answer choice D is getting at.

30. **A.** This patient is showing signs of normal aging and does not require further evaluation. Certain memory performances on cognitive testing, like procedural, primary, and semantic memory, are well-preserved with age. Skills, ability, and knowledge that are overlearned, well-practiced, and familiar, like vocabulary or general knowledge, remain stable or improve up to 0.2 standard deviations per decade through the seventh decades, but even these processes can begin to decrease with further aging. The ability to recognize familiar objects and faces, as well as to maintain appropriate visual perception of objects, remains stable over the lifetime. Episodic and working memory and executive function are the specific domains of cognition most affected by "normal" aging. These are late-life

changes, occurring after the sixth decade and have a linear or accelerating decline with further aging. Processing speed decreases with age and can have a global effect on the testing performance of other neurocognitive domains in any timed test.

Executive function is critical to engagement in purposeful, independent, and self-preserving behavior and is necessary for an older person to successfully manage their own medical illnesses. Executive function declines with age, and more dramatically after age 70. Attention span decreases with even simple attentive tasks. In particular, there is decrease in the ability to focus on a task in a busy environment and ability to perform multiple tasks at one time. Problem-solving, reasoning about unfamiliar things, processing and learning new information, and attending to and manipulating one's environment show a steady decline (by about -0.02 standard deviations per year) after peaking around age 30. Language abilities (verbal fluency and the ability to name objects) demonstrate some late-life decline, particularly after age 70.

31. **C.** Both GERD and peptic ulcer disease (PUD) are in the differential diagnosis for this patient's symptoms. The diagnosis of GERD is based on clinical symptoms alone with esophageal testing reserved for refractory cases. *H. pylori* infection is common and should be ruled out in a patient with symptoms suspicious for PUD. The test is also non-invasive, and the consequences of untreated PUD can be grave.

32. **D.** ACE inhibitors are always beneficial in MI with EF < 40%.

33. **D.** In patients near a threshold for treatment based on total CV risk and in patients above a threshold for treatment, repeating measurements every three years is recommended. According to many community practice standards of care such follow-up studies are performed yearly.

34. **D.** Fecal impaction is the likely diagnosis, especially as constipation is a common side effect of the oxybutynin the patient is taking for urinary incontinence. Oxybutynin blocks not only the muscarinic M₃ receptor subtypes, but also the M₁ receptor subtypes. This action accounts for the common adverse effects associated with oxybutynin: new-onset constipation, dry mouth, flushing, and heat intolerance.

35. **A.** Both overflow incontinence and benign prostatic hyperplasia (BPH) are in the differential diagnosis for this patient's symptoms. Overflow incontinence typically presents with continuous urinary leakage or dribbling in the setting of incomplete bladder emptying. Associated symptoms can include weak or intermittent urinary stream, hesitancy, frequency, and nocturia. When the bladder is very full, stress leakage can occur or low-amplitude bladder contractions can be triggered resulting in symptoms similar to stress or urgency incontinence.

BPH is a histologic diagnosis that becomes more prevalent with age, although some men with BPH are asymptomatic. Approximately 50 percent of men at age 50 and up to 80 percent of men at age 80 have lower urinary tract symptoms (LUTS) attributable to BPH. Common manifestations include storage symptoms (increased daytime frequency, nocturia, urgency, and urinary incontinence) and voiding symptoms (slow urinary stream, splitting or spraying of the urinary stream, intermittent urinary stream, hesitancy, straining to void, and terminal dribbling). Post-void residual urine volume determination is useful in men with evidence of urinary obstruction or suspected neurologic involvement of the genitourinary tract and prior to initiation of an anticholinergic drug (which may decrease bladder contractility). Normal men have less than 12 mL of residual urine, but most urologists are not concerned unless the post-void residual volume is greater than 100 to 200 mL. In addition to being a possible indicator of BPH, a large residual volume is probably associated with increased risk of infection and is a precursor to bladder decompensation.

36. **C.** This patient most likely has a scaphoid fracture. Scaphoid fractures are the most common carpal bone fracture and typically occur from a fall onto an outstretched arm with the wrist in dorsiflexion. Suspect a scaphoid fracture in any patient with wrist pain following a fall. When a definitive diagnosis cannot be

determined at presentation and a scaphoid fracture is suspected on clinical grounds, even if radiographs are negative, the patient should be placed in a volar wrist splint or preferably a thumb spica splint or cast until a definitive imaging study can be performed.

37. **E.** This patient has diminished renal function as indicated by her elevated creatinine, which according to the Cockcroft-Gault equation is approximately a GFR of 38. The US Food and Drug Administration (FDA) revised its labeling of metformin, which previously had identified metformin as contraindicated in women and men with serum creatinine levels ≥ 1.4 mg/dL (124 micromol/L) and ≥ 1.5 mg/dL (133 micromol/L), respectively. The use of metformin is contraindicated in patients with an eGFR < 30 mL/min, and the initiation of metformin is not recommended in patients with an eGFR between 30 and 45 mL/min. For patients taking metformin whose eGFR falls below 45 mL/min, the benefits and risk of continuing treatment should be assessed, whereas metformin should be discontinued if the eGFR falls below 30 mL/min. Thus while metformin is not absolutely contraindicated in this patient, this is the best of the answer choices presented. It is possible also that this question was written based on previous guidelines that contraindicated metformin at lesser degrees of renal impairment.

38. **D.** A physician cannot revoke a person's driver's license (only the department of motor vehicles can do that). But the family can ensure that the patient does not have access to car keys until a formal evaluation can be completed.

39. **A.** This patient likely has a viral pharyngitis without meeting the criteria for strep pharyngitis. He does not have a fever, tonsillar exudates, tender cervical lymphadenopathy, and he has a cough. Viral pharyngitis requires only supportive treatment, including use of warm saline gargles. A patient with a very high Centor score would be treated for strep pharyngitis without need for a confirmatory test, a patient with an equivocal score should be given a rapid strep test, and any patient testing negative should be given a throat culture in order to prevent rheumatic heart disease in case of a false negative. Infectious mononucleosis should be suspected in patients 10 to 30 years of age who present with sore throat and significant fatigue, palatal petechiae, posterior cervical or auricular adenopathy, marked adenopathy, or inguinal adenopathy. An atypical lymphocytosis of at least 20 percent or atypical lymphocytosis of at least 10 percent plus lymphocytosis of at least 50 percent strongly supports the diagnosis, as does a positive heterophile antibody test. This patient's clinical picture is not suspicious enough for mono to warrant a monospot test.

40. **A.** This patient has a macrocytic anemia, which is commonly caused by either a folate or vitamin B₁₂ deficiency. It takes years to develop vitamin B₁₂ deficiency versus a few months to develop folate deficiency.

41. **C.** Individuals with suspected iron deficiency should have laboratory testing that is stratified for their age and possible related conditions. For infants without risk factors for (or evidence of) lead toxicity, a complete blood count may be sufficient. For children and young adults, it is appropriate to obtain a reticulocyte count, review the RBC indices and/or the blood smear, and test the stool for occult blood. For individuals with other medical conditions, it is appropriate to measure iron studies (serum ferritin, total iron binding capacity [TIBC]), and serum iron. This 55 year old woman should have her ferritin checked to determine whether she has iron deficiency anemia likely due to poor diet or occult blood loss.

42. **B.** The most common cause of iron deficiency anemia in the elderly is GI blood loss. The physician started with the best initial/least invasive therapy, a trial of iron supplementation. Her lack of response indicates that there is ongoing blood loss. Thus the best next step is a colonoscopy. Other causes for lack of improvement with iron supplementation (in addition to ongoing blood loss) could be nonadherence to iron or celiac disease.

43. **A.** It is not certain whether she is telling the truth. Thus the physician should speak to both at the same time in order to assess the validity of his concerns.

44. **C.** This patient has elevated free T₄, symptoms of hyperthyroidism, and neck tenderness with decreased uptake on RAIU scan, which indicates subacute thyroiditis. In subacute thyroiditis the thyroid gland is typically slightly or moderately diffusely or asymmetrically enlarged, and nearly always tender. In some

cases, the pain is so severe that the patient cannot tolerate palpation of the neck. Both thyroid lobes are involved from the beginning in most patients, but the pain, tenderness, and enlargement can be unilateral or start on one side and later spread to the other side days or even weeks (so called “creeping thyroiditis”) later. Temperature elevations also can occur. Subacute thyroiditis is fundamentally a clinical diagnosis. In most patients, clinical manifestations (the presence of neck pain, often radiating upward to the jaw, marked thyroid tenderness, and a diffuse goiter) are sufficient to establish the diagnosis. Symptoms and signs of hyperthyroidism may or may not be present, but the serum thyroid-stimulating hormone (TSH) is usually suppressed (typically <0.1 mU/L) and free thyroxine (T4) and triiodothyronine (T3) concentrations elevated, particularly in the early stages of the illness. Serum TSH, free T4, and T3 should be measured in all patients in whom there is a clinical suspicion of subacute thyroiditis. We also typically measure an erythrocyte sedimentation rate or C-reactive protein (CRP) level and obtain a radioiodine or technetium imaging study. A high erythrocyte sedimentation rate and/or CRP measurement and a low radioiodine uptake (usually less than 1 to 3 percent) during the hyperthyroid phase help confirm the diagnosis.

45. **A.** This is a bad question, as the CDC recommends PPSV23 for everyone 2 years of age and older at increased risk of disease, which explicitly includes those who smoke cigarettes or have asthma. Thus the correct answer is C, he should receive the flu vaccine and the PPSV23 vaccine, but because he has completed all of his childhood vaccinations, which include Hib, he does not need another Hib vaccine.

46. **D.** This patient with lower back injury and symptoms of radiculopathy likely has either a muscle strain or a disc herniation, and should be initially managed with NSAIDs.

47. **C.** This is a clinically depressed patient who should be screened for suicide.

48. **D.** For an overweight person, the greatest decrease in SBP occurs with at least 10% weight loss.

49. **D.** Food stuck in a patient’s throat with a 20 year history of GERD that has not improved with PPIs should prompt an EGD to rule out complications of long term GERD, for example Barrett’s esophagus, adenocarcinoma, or esophageal stricture.

50. **D.**